

CCT College Dublin
Assessment Cover Page

Module Title:	Strategic Business Information Technology Problem Solving for Industry (Capstone Project)
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Declaration

By submitting this assessment, I confirm that I have read the CCT policy on Academic Misconduct and understand the implications of submitting work that is not my own or does not appropriately reference material taken from a third party or other source. I declare it to be my own work and that all material from third parties has been appropriately referenced. I further confirm that this work has not previously been submitted for assessment by myself or someone else in CCT College Dublin or any other higher education institution.

Optimizing Ready Meal Stock Management at Lotts & Co. Using Data Analytics

GitHub: <https://github.com/MorganeTassoneiro/CA1---Project-Strategic-Analysis.git>

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1. INTRODUCTION

Project Goal: Lotts & Co faces two challenges: lost sales due the stockouts and waste due to overstock. My solution is to use data analytics to address these issues. By minimizing waste and maximising stock availability the store increases revenue and customer loyalty.

Why This Matters: Stock shortages lead to lost sales opportunities and dissatisfied customers, while overstock leads to waste, increased storage and financial loss. Ready meals have a limited shelf life, so good inventory management is crucial to minimize food waste and maximise sales efficiency.

2. BUSINESS CASE FOR LOTTS & CO.

Improving stock management could raise revenue, reduce overhead and increase competitive edge. The ready-meal market is expected to grow 5.1% annually from 2022 to 2030 (Consumer Goods, 2020). Ready meals are cost-effective and take less time to prepare being precooked for those who have a busy lifestyle or own schedules.

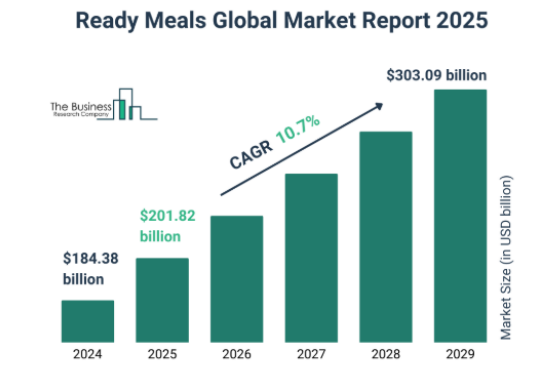


Figure 1: Ready Meals Global Market Report 2025 (The Business Research Company, 2025)

Commercial Relevance

- Who will use the system? Store managers for informed decisions about stock levels and orders. Staff often use manual checks which may not always lead to the best outcome and can cause effectiveness problems.
- Competitive Advantage: Prevents lost sales, improves customer satisfaction, and reduces storage costs. Cutting overstock frees up storage for high-demand items.

3. BUSINESS ANALYSIS METHOD: SWOT ANALYSIS

1. Strengths

- An extreme demand for premium ready meals allows for maintaining higher prices and a strong brand image.
- Ensuring consistent availability from suppliers, reducing the change of stockout products.
- Attracting loyal customers who trust the brand quality, which leads to repeated purchases and word-of-mouth recommendations.

2. Weaknesses

- Frequent unavailability of popular ready meals leads to lost sales and disappointed customers. (Corsten & Gruen, 2004)

- Slow-moving meals cause overstock resulting in waste and financial loss, sales analytics can identify which meals don't sell quickly and allow managers to adjust orders and reduce waste. (Czuczman, 2024)

3. Opportunities

- Data analytics can be used to ensure the right meals are in stock, analysing historical sales can be aligned with the actual demand.
- Real-time sales to align stock using this technology enables rapid decision-making, supporting special promotions or emergency restocking. This can respond to market trends faster

4. Threats

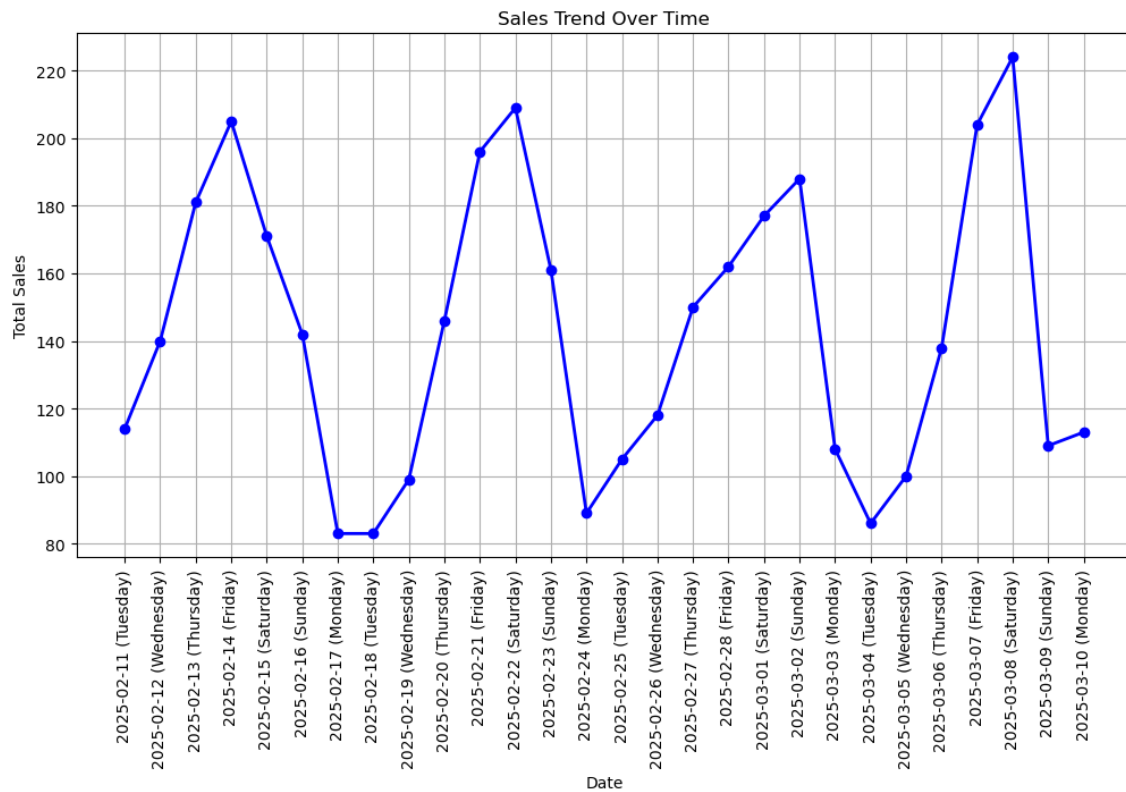
- Inventory systems of competitors being more efficient. Implementing advanced analytics ensures the store keeps pace or surpasses competitors
- Delays from suppliers keep stock unavailability especially if a popular meal is frequently missing.

5. DATA COLLECTION & ANALYSIS

- ✓ Period Collected: 10/02 to 10/03 (one month).
- ✓ Dataset Size: 8 columns, 1035 rows I captured daily sales and stock information.
- ✓ Goal: Identify demand trends, and bestselling items and reduce waste from overstock using Python.

Sales Trend Over Time:

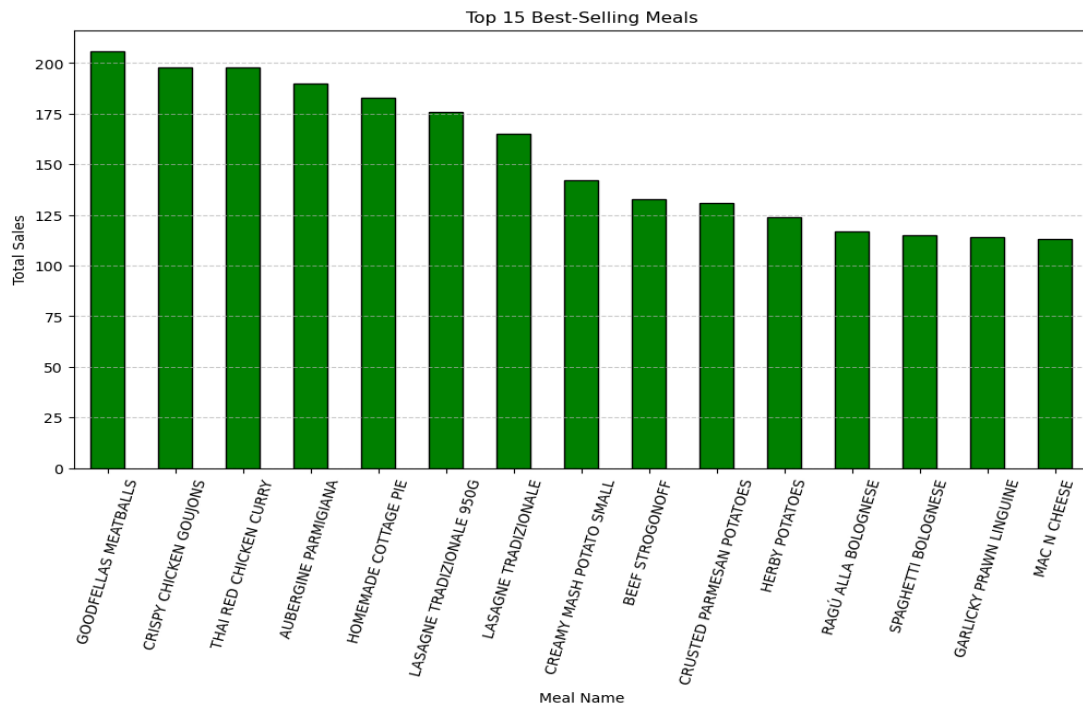
The graph shows the sales peak are on Fridays and Saturdays and the lowest on Mondays and Tuesdays



- **Strategic impact:** This suggests that stock replenishments should be scheduled before these peak days to avoid shortages

Best-Selling Meals:

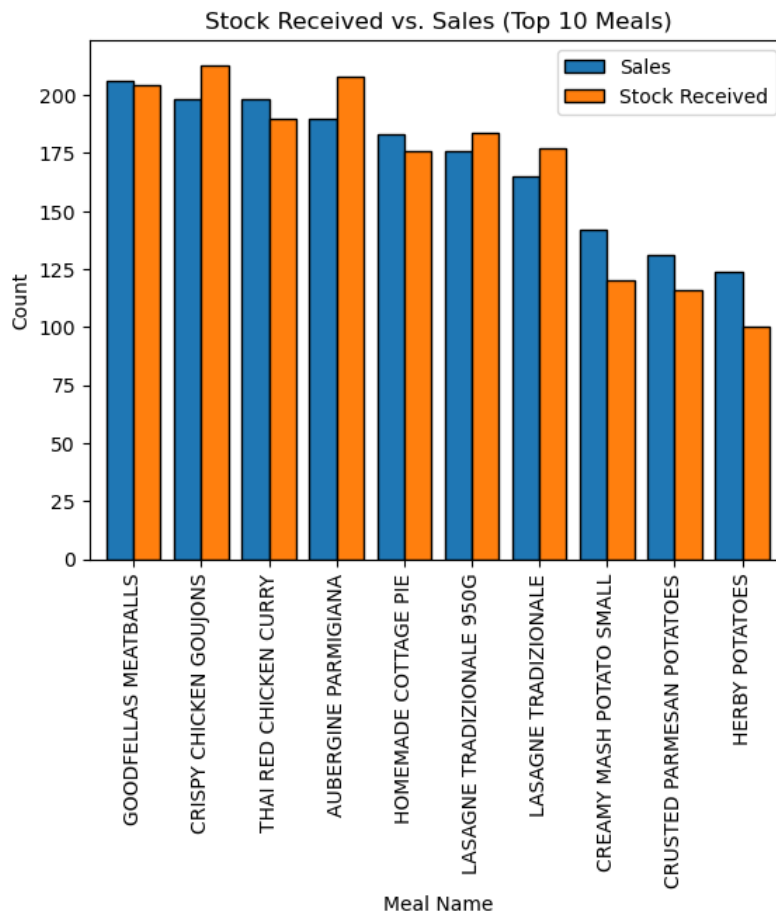
Meatballs, Thai Curry, Crispy Chicken Goujons, Aubergine Parmigiana, and Homemade Cottage Pie are ranked highest.



- **Strategic impact:** This item should be prioritized to avoid disappointed customers or losing revenue

Stock Needs vs. Sales

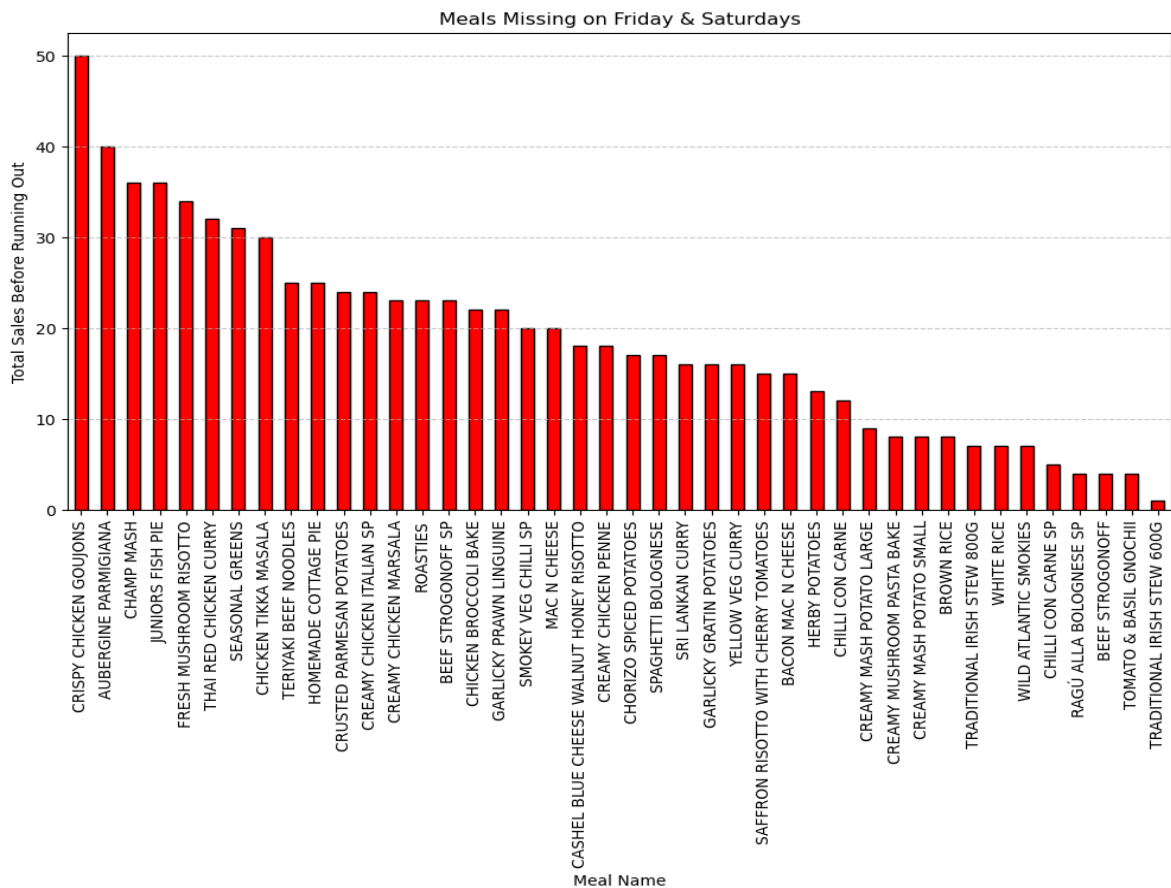
Some meals example Meatballs and Thai Curry, have a higher demand than the stock received causing stockout, whereas lasagna tradizionale and aubergine parmigiana are often overstocked leading to waste.



- **Strategic impact:** A data-driven is needed to balance supply with actual demand.

Meals that are missing on Fridays and Saturdays (busiest days)

Popular items are being sold out on Fridays and Saturdays due to no replenishment after Thursday.



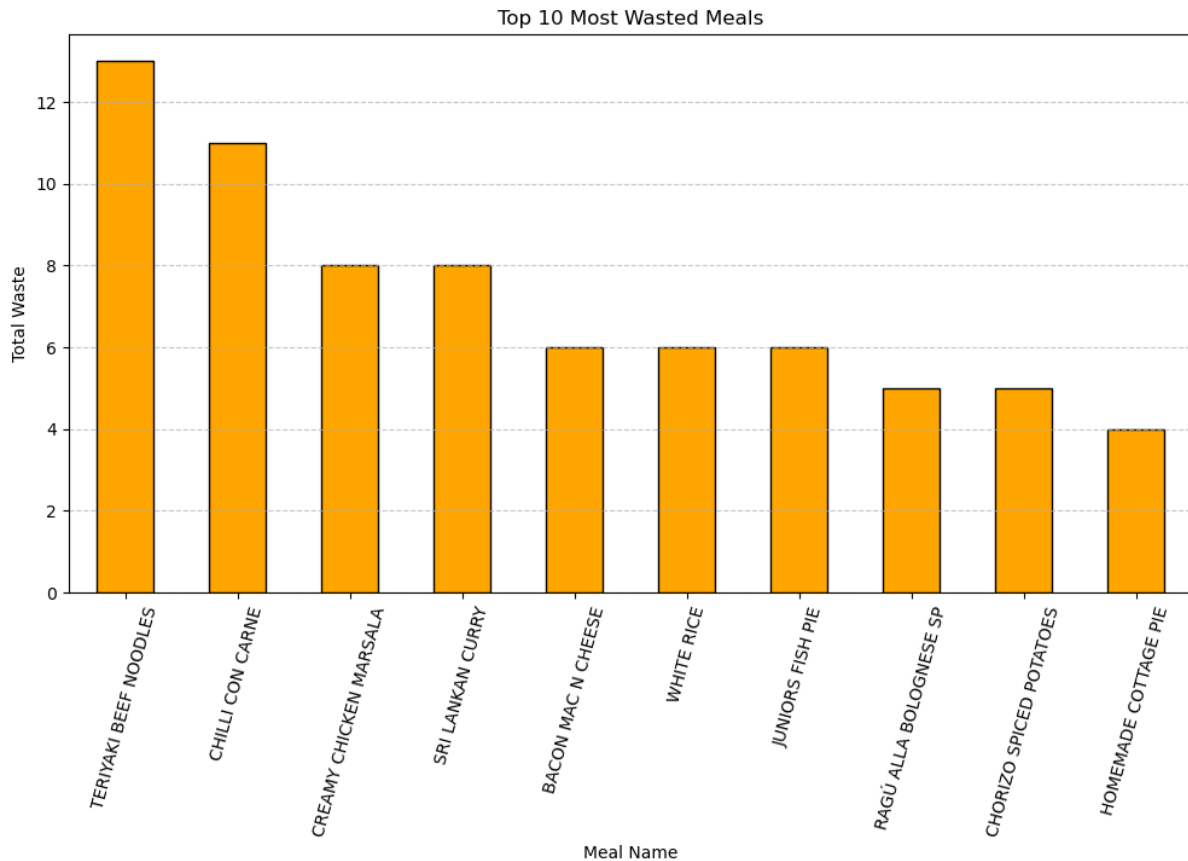
Meals Missing on Friday & Saturdays:

	Date	MealName	Sales
138	2025-02-14	AUBERGINE PARMIGIANA	11
142	2025-02-14	CHICKEN BROCCOLI BAKE	6
143	2025-02-14	CHICKEN TIKKA MASALA	6
146	2025-02-14	CREAMY CHICKEN ITALIAN SP	3
147	2025-02-14	CREAMY CHICKEN MARSALA	3
..	...	...	...
947	2025-03-08	CHAMP MASH	6
948	2025-03-08	CHORIZO SPICED POTATOES	2
950	2025-03-08	CREAMY MASH POTATO SMALL	8
951	2025-03-08	CRUSTED PARMESAN POTATOES	9
957	2025-03-08	WHITE RICE	2

- **Strategic Impact:** Introducing a better weekend delivery to buffer stock and ensure availability during peak demands

Stock waste analyses

Meals such as Teriyaki Beef Noodles and Chili con carne show overstock and end up being wasted.



- **Strategic Impact:** Adjusting order quantities or removing slow-moving items reduces financial loss and storage costs.

6. TECHNOLOGY SELECTION

Chosen Technology: Python (Pandas & Matplotlib)

- Python is a free tool and has no license fees, making it scalable for future growth without an additional cost
- There is a vast ecosystem of libraries (for example: Panda, Numpy) that will help with everything from data cleaning to predictive modelling. That means fast troubleshooting and improvement. (Python, 2024)
- Python can be customized to match the business needs for example: the models for peak days at Lotts& Co

Alternative: Excel

- Excel has advanced features in Microsoft 365 that require a paid subscription
- Excel will handle basic operations and small datasets and probably will struggle with large-scale data, which can be critical for a growth company such as Lotts and Co. (Gomes, 2024)

- Integration with other solutions can result in extra costs if the business needs more advanced features later

7. LEGAL & ETHICAL CONSIDERATIONS FOR READY MEALS

Legal Considerations:

- FSAI Regulations on Food Safety: Ready meals should conform to safety norms regarding storage conditions, handling, and expiry tracking (Food Safety Standards & Regulations, 2024).
- EU Food Information to Consumers Regulation No 1169/2011: The labelling of nutritional information, allergens, and expiry dates must be accurate (European Commission, 2024).
- Waste Disposal Compliance: The proper disposal and recycling of food waste are governed by the European Union Waste Framework Directive 2008/98/EC to minimize the impact on the environment (European Environment Agency (EEA), 2023).
Competitive Advantage for Lotts & Co.
- No personal data is collected; only aggregated sales figures are used.

Food Waste Reduction & Sustainability

- Minimizing Food Waste: By reducing over-ordering through utilization of analytics, stock control for instance is promoted to the sustainable management of them (WRAP UK, 2023)

8. COMPETITIVE ADVANTAGE FOR LOTTS & CO.

- Reduce waste: with precise demand the means orders will be made as they sell, cutting extra expenses and freeing up storage space for higher margin items.
- Sustainability: As waste decreases Lotts can project a more environment-conscious brand, increasing customers who value ethical business practice
- Consistent Availability of Top Sellers: Keeping the best sellers on peak days (Friday & Saturday) ensures a reliable shopping experience and recommendations.
- Customer Loyalty: By meeting demand and reliably the store will differentiate itself from competitors, retaining and attracting clientele.

9. CONCLUSION

This analysis shows a clear market opportunity to improve the ready-meal stock management at Lotts & Co. The SWOT analysis pointed out that stockouts and overstock are the main problems impeding sales and customer loyalty. By leveraging data analytics—with the support of open-source tools such as Python the company can align supplies to meet customer demand, minimizing waste while also preventing lost sales. Legitimate and ethical considerations (food safety, labelling, environmental regulations, etc.) ensure the store's compliance and build a positive image as a sustainable and customer-friendly business.

10. RECOMMENDATIONS

Forecasting Weekly Demands:

- Use historical sales data and Python analysis libraries, can be used to predict the following week's demand, especially on Friday and Saturday.
- Adjust delivery schedules and stock levels to avoid shortages or waste

Training Session

- Have brief training sessions for managers/relevant staff in interpreting daily dashboards on sales and acting on forecasts

Waste & Customer Feedback

- Data analytics are to be used on slow-moving products
- Customer feedback asking whether an item was in stock will help refine the forecasting model.

Look-Back/Review

- Review, if necessary, supplier contracts to allow for more frequent or flexible deliveries.
- A monthly review of performance metrics (waste, stockouts, sales trend).

By implementing this Lotts&Co can maintain a competitive edge, enhance customer satisfaction have a more efficient and sustainable inventory management

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